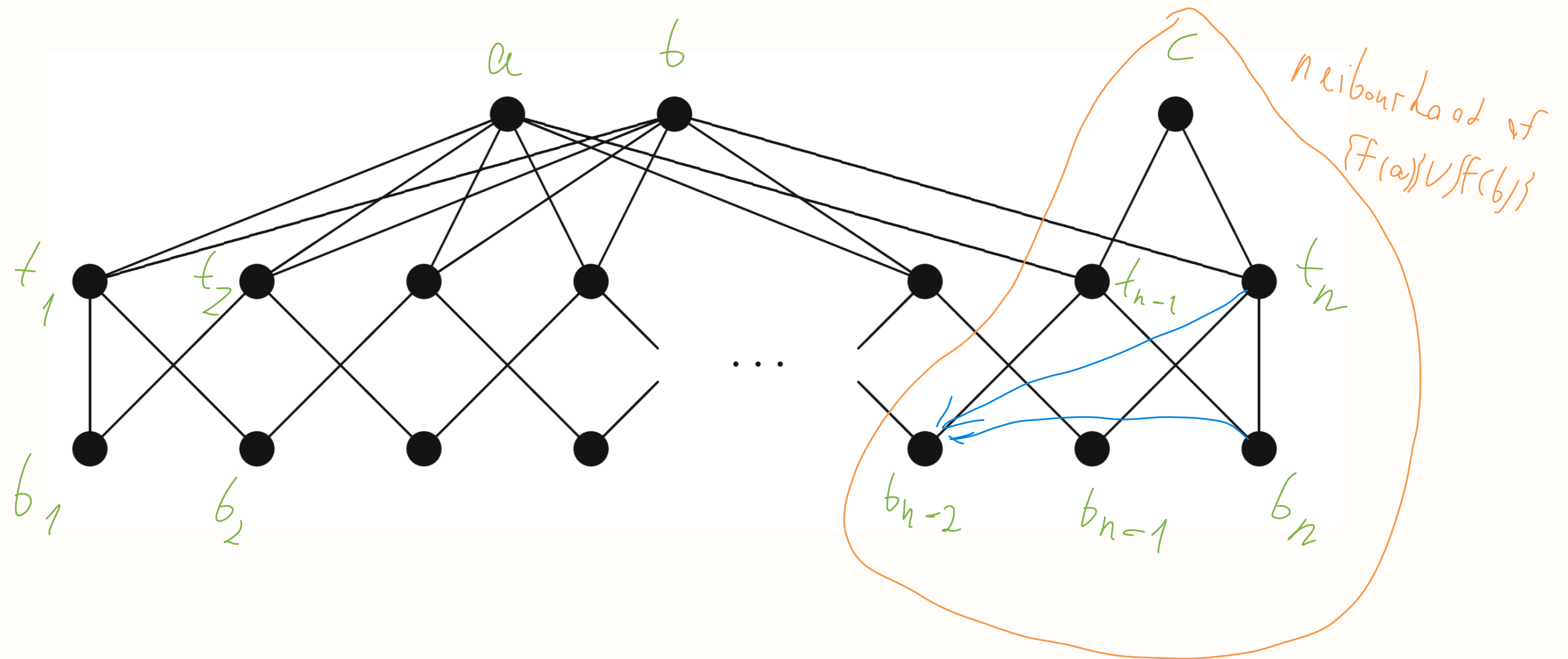


2-29a

Prove this class has FPP



Let's try similar to 1-1sa

by criterion we have f is FPF $\Rightarrow f(b) \neq b \Rightarrow$

$$\Rightarrow f(b) = \begin{bmatrix} c \\ t_{n-1} \end{bmatrix}$$

same way $f(a) = \begin{bmatrix} c \\ t_n \end{bmatrix}$

In any case everything except $c \subseteq$

$$\subseteq \{b_{n-2}, b_{n-1}, b_n, c, t_{n-1}, t_n\} \text{ (call it } A)$$

$$f(t_{n-1}) \in A \Rightarrow f(t_{n-1}) = \begin{bmatrix} b_{n-1} \\ t_n \end{bmatrix}$$

$$f(t_n) = \begin{bmatrix} t_{n-1} \\ b_{n-2} \end{bmatrix}$$

$$f(b_{n-2}) = \begin{bmatrix} b_{n-1} \\ b_n \\ t_n \end{bmatrix}$$

$$f(b_{n-1}) = \begin{bmatrix} b_{n-2} \\ b_n \\ t_{n-1} \end{bmatrix}$$

$$f(b_n) = \begin{bmatrix} b_{n-1} \\ b_{n-2} \end{bmatrix}$$

Let's consider cases of

(b_n, t_n)

1) (b_{n-1}, t_{n-1}) - contradicts $f(b_n) \leq t(t_n)$

2) (b_{n-2}, b_{n-2}) - no options for $f(t_{n-1})$ case $t(t_{n-1}) \geq f(b_n) = b_{n-2}$

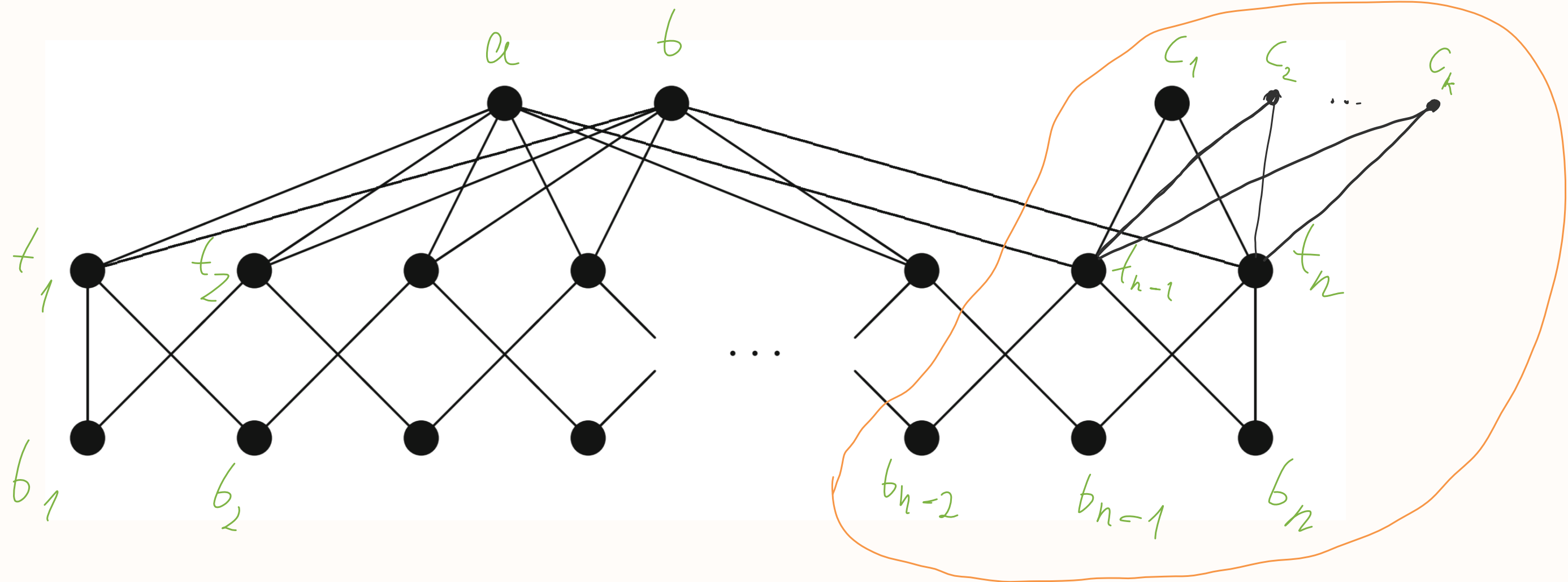
3) (b_{n-1}, b_{n-2}) - contradicts $t(b_n) \leq f(t_n)$

4) (b_{n-2}, t_{n-1}) - $t_{n-1} \geq b_n \Rightarrow f(t_{n-1}) \geq f(b_n) = b_{n-2}$
either t_n or b_{n-1}

- no options

2-296

for example adding new "c"



Neighborhood of $\{f(a)\} \cup \{f(b)\}$

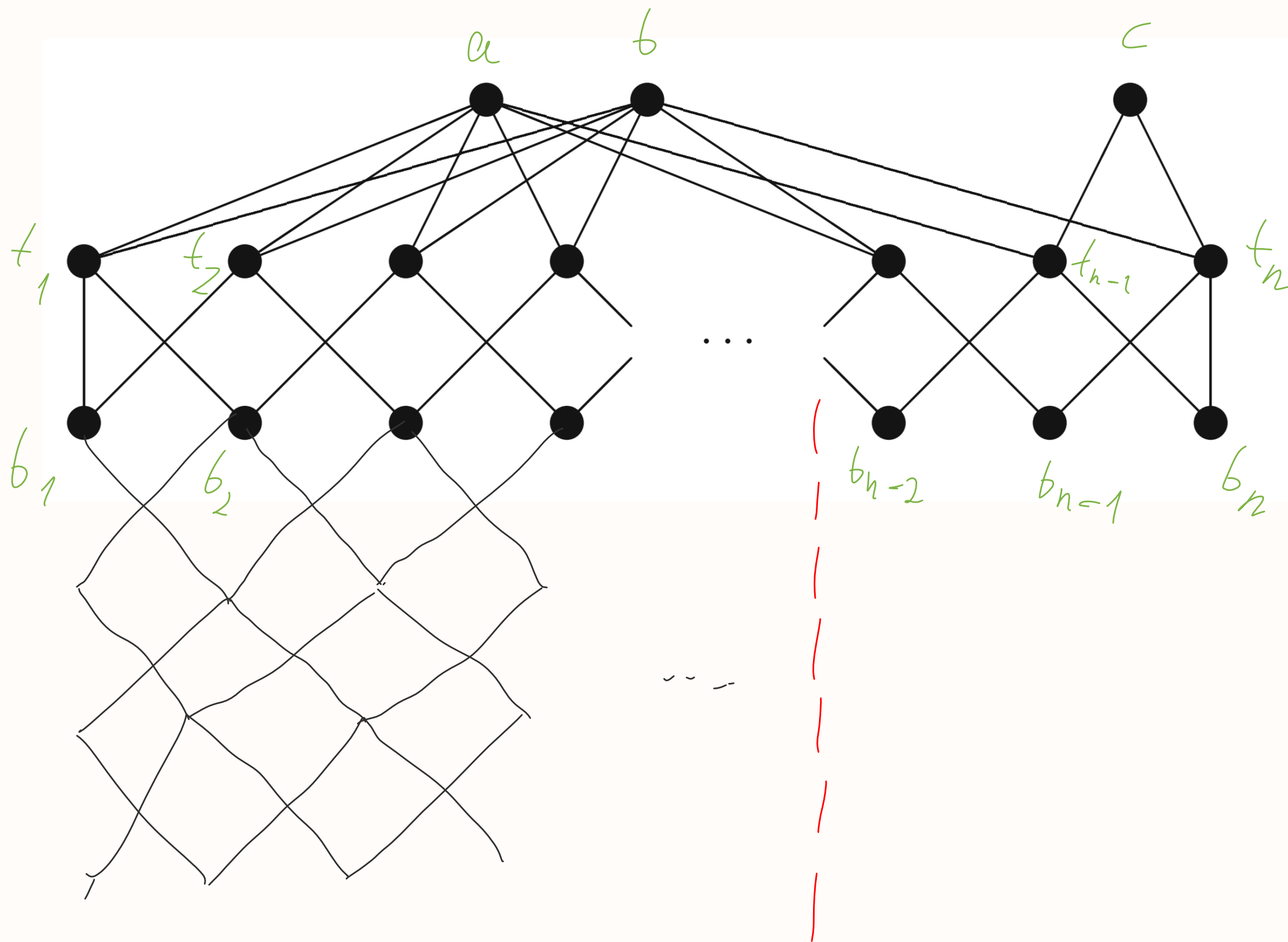
Will the same proof work? Yes:

the options for $f(a)$ are now t_n and $\{c_k\}$
—— || ——— $f(b) = t_{n-1}$ and $\{c_k\}$ $\Rightarrow$ same options for

$f(b_{n-2}), f(b_{n-1}), f(b_n), f(t_{n-1})$ and $f(t_n)$ because all of them
are $\sim$ to all c_i ($\forall 1 \leq i \leq k$) $\Rightarrow$

$\Rightarrow$ New order has FPP

As a second generalization we can add crown tower
on the strict left of b_{n-2}



No new elements $\sim b_{n-2}!$ — same proof would work